



An individualized gait pattern prediction model based on the least absolute shrinkage and selection operator regression

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ABSTRACT

Gait pattern prediction is an essential function of individualized motion control of the lower-limb exoskeleton. This paper presents a novel gait pattern prediction model based on the least absolute shrinkage and selection operator (LASSO) regression. Gait data were collected from one hundred and twenty healthy young adults (78 males and 42 females), who were instructed to walk back and forth on a 12-meter-long walking platform while having their heel strike coinciding with the beat of a metronome. The lower-limb joint (i.e., the hip, knee and ankle) angular kinematics were segmented, resampled and transformed into Fourier coefficients. The LASSO regression model with age, gender and 14 anthropometric parameters as prediction variables was trained and used to estimate the Fourier coefficients which were then applied in the lower-limb joint angle trajectory reconstruction. The results showed that the root mean square errors between the actual and predicted joint angle trajectories ranged from 3.41° to 4.55°. The parameters of the linear fit method further revealed the waveform similarity between the actual and predicted joint angle time series. These results suggested that the proposed model was able to accurately predict lower-limb joint kinematics during gait. Application of the proposed model can help resolve the overfitting problem, and provides a new solution to individualized gait pattern prediction.

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1. Introduction

Lower-limb exoskeletons have gained increasing attention from gait rehabilitation researchers and practitioners (Dollar and Herr, 2008; Schwartz and Meiner, 2015). The control of the lower-limb exoskeleton motions is dependent on pre-planned gait patterns (Jimenez-Fabian and Verlinden, 2012). Traditionally, pre-planned gait patterns were obtained by using the gait data recorded from able-bodied persons (Vallery et al., 2009). A limitation of such traditional approach is that it cannot address individual differences. As a result, it might lead to discomfort or even injuries for wearers.

Recently, a few individualized gait pattern prediction models have been proposed for exoskeleton motion planning (Koopman et al., 2014; Luu et al., 2014; Yun et al., 2014; Wu et al., 2018; Ferreira et al., 2018). For example, Koopman et al. (2014) proposed a multi-variable regression model to predict hip and knee joint kinematics, in which the joint trajectories were reconstructed by piece-wise quantile spline fitting. Yun et al. (2014) developed a stochastic mapping function model based on the Gaussian process

regression (GPR) for lower-limb joint trajectory prediction. The input of their model included 14 individual-specific parameters, and the output was the probability distribution of lower-limb joint displacement and angles. Similarly, Wu et al. (2018) also implemented the GPR to predict the individualized gait pattern.

A major limitation of the existing individualized gait pattern prediction models is that they typically included a relatively large number of prediction variables representing different aspects of individual differences (Horst et al., 2017). This can result in the overfitting problem (Krzanowski, 1998), which means that the models might well fit the training data, but fail to predict gait patterns beyond the training dataset. In addition, the large number of prediction variables would increase model variance and compromise model's generalizability (Ferber et al., 2016).

To overcome above-mentioned challenges, this study proposed a novel individualized gait pattern prediction model based on the least absolute shrinkage and selection operator (LASSO) regression. The LASSO regression is a regression analysis method that penalizes the magnitude of the coefficients of prediction variables by imposing a constraint during parameter estimation. This process is known as the "shrinkage" process (Tibshirani, 1996). The regression coefficients of the identified irrelevant prediction variables

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can even be reduced to zero. Thus, the LASSO regression can help reduce the number of prediction variables, and minimize the possibility of model overfitting.

2. Method

2.1. Experiment

One hundred and twenty healthy young participants including 78 males and 42 females were involved in the experiment. All participants were self-reported without any medical conditions that would affect their normal gait patterns. They all signed an informed consent form before the experiment. This study was approved by the Institutional Review Board of Shenzhen University. The participants' demographic and anthropometric data were presented in Table 1.

An eight-camera motion capture system (VICON, Oxford Metrics, Oxford, UK) was used to capture gait kinematics with the sampling frequency at 200 Hz. Thirty-nine reflective markers were placed on the bony landmarks according to VICON's Full Body Plug-in-Gait Model. Participants were asked to walk back and forth on a 12-meter-long walking platform while having their heel strike coinciding with the beat of a metronome. The frequencies of the metronome were set in the way corresponding to six cadence levels, i.e., 36 steps/minute, 48 steps/minute, 60 steps/minute, 72 steps/minute, 84 steps/minute, and 96 steps/minute. The selected cadence levels were all below the normal walking cadence level because gait pattern prediction is mainly for frail people who need rehabilitation. Each participant conducted one three-minute walking trial under each of the six cadence levels. The cadence levels were presented in a random order across participants. A two-minute break was given between two consecutive trails to avoid confounding fatigue effects.

2.2. Gait pattern prediction model

Age, gender and 14 anthropometric parameters were included as the prediction variables (Please find the 14 anthropometric parameters in Table 1). To make the prediction variables comparable with each other, age and anthropometric parameters were normalized as follows:

$$\bar{x} = \frac{x - \text{Min}}{\text{Max} - \text{Min}} \quad (1)$$

where x represented the original magnitude of the parameter; Max and Min represented the maximum and minimum values of the parameter among all participants. The normalized parameter ($\bar{x}$)

Table 1
Prediction variables.

Prediction variables	Mean $\pm$ SD	Range
X1 Age (year)	22.2 $\pm$ 2.3	18–31
X2 Gender	N.A.	N.A.
X3 Height (mm)	1681.1 $\pm$ 78.6	1500–1860
X4 Weight (kg)	58.3 $\pm$ 9.6	40–90
X5 Shoulder width (mm)	321.5 $\pm$ 28.9	257–395
X6 Trunk length (mm)	466.3 $\pm$ 41.3	365–570
X7 Hip width (mm)	266.3 $\pm$ 27.4	193–350
X8 Bi-trochanteric width (mm)	312.7 $\pm$ 24.5	245–400
X9 Lower-limb length (mm)	868.2 $\pm$ 54.1	736–1070
X10 Thigh length (mm)	473.3 $\pm$ 30.4	400–555
X11 Knee width (mm)	96.8 $\pm$ 8.3	79.3–123.6
X12 Calf length (mm)	395.6 $\pm$ 26.7	325–475
X13 Malleolus width (mm)	66.9 $\pm$ 4.7	55.6–78.2
X14 Malleolus height (mm)	70.8 $\pm$ 7.8	40.9–99.0
X15 Foot width (mm)	239.3 $\pm$ 15.9	206–274.5
X16 Foot length (mm)	90.8 $\pm$ 7.2	72–110

ranged from 0 to 1. Gender, which is a nominal variable, was converted into the numerical dummy variable by one hot encoder, where female = 0 and male = 1 (Bullen et al., 1997). The selected prediction variables formed a 1x16 vector $X = (X_1^i, X_2^i, \dots, X_{16}^i)^T$, which was used as the input for the proposed gait pattern prediction model.

The lower-limb joint angle trajectories were segmented using the “maxima-to-maxima” rule, where the local maxima in the angle trajectory waveform were identified by using the “peakdet” function in Python (Billauer, 2012), and were further used to segment the cyclic data. Twenty segmented “gait cycles” were randomly selected from each walking trial. The segmented data were not in the same length due to inter-personal and intra-personal variations. Thus, the data from each gait cycle were resampled by a near-neighbour interpolation to unify the data length under each cadence. To avoid over-distortion in time, the numbers of data points retained in each gait cycle were 666, 500, 400, 333, 285, and 250, respectively, for the cadence at 36 steps/minute, 48 steps/minute, 60 steps/minute, 72 steps/minute, 84 steps/minute, and 96 steps/minute (given that VICON sampling frequency was 200 Hz). After data resampling, the average joint angle trajectories were obtained for each participant by taking the means across the 20 gait cycles. The time series of the average joint angle trajectories were then decomposed by a 5th order Fourier Transformation (FT) following the method described in Luu et al. (2014), as follows:

$$f(t)^i = \frac{1}{2}c_0^i + \sum_{n=1}^5 (c_n^i \cos n\omega t + \phi_n^i) \quad (2)$$

where $f(t)^i$ represented the time series of the i^{th} lower-limb joint angle trajectory (i : 1 = hip, 2 = knee, 3 = ankle); c_0^i , c_n^i and ϕ_n^i ($n = 5$) were the Fourier coefficients. These Fourier coefficients formed a 1x11 vector $Y = (c_0^i, c_1^i, c_2^i, \dots, c_5^i, \phi_1^i, \phi_2^i, \dots, \phi_5^i)^T$, which were considered as the output of the LASSO regression model.

Data from 63 male participants and 34 female participants (at least 80% of data from each gender) were randomly assigned to the training dataset, and the rest data were assigned to the testing dataset. The training data were used to train the model and estimate the model coefficients. The test data were used for model evaluation.

The training data were fit in a multiple multivariant regression model as:

$$Y = \beta^{\text{LASSO}} X \quad (3)$$

where $\beta^{\text{LASSO}} = (\beta_1^i, \beta_2^i, \dots, \beta_{16}^i)^T$ were the LASSO coefficients.

The LASSO adds a penalty equivalent to absolute value of the magnitude of coefficients into the objective function that determines the value of $\hat{\beta}^{\text{LASSO}}$ (also known as the $L1$ penalty), which can be written in the mathematic form as:

$$\hat{\beta}^{\text{LASSO}} = \text{argmin} \left\{ \sum_{i=1}^N \left(Y_i - \sum_{j=1}^M \hat{\beta}_j^{\text{LASSO}} X_{ij} \right)^2 + \lambda \sum_{j=1}^M |\hat{\beta}_j^{\text{LASSO}}| \right\} \quad (4)$$

where N represents the number of observations (i.e., the sample size in the training data), and M represents the number of prediction variables. $Y_i \in Y$, $X_{ij} \in X$. λ is known as the regularization parameter, which controls the strength of the penalty, and the shrinkage of model coefficients (β^{LASSO}). The larger the value of λ , the greater the amount of shrinkage. λ and β^{LASSO} in the present study were determined by a 10-fold cross validation with the training dataset. During each training iteration, $\hat{\beta}^{\text{LASSO}}$ were estimated by a cyclic descent coordinate optimization of the objective function (Wu and Lange, 2008). During this optimization procedure, the $L1$ penalty

in the objective function can force the β^{LASSO} of irrelevant prediction variables to be zero. The above described model development was realized by implementing a customized script written based on the *class.sklearn.linear_model.Lasso* from Scikit-learn Python module (v0.21.2) (Pedregosa et al., 2011). The predicted Fourier coefficients were then used to reconstruct the joint angle trajectories by the inverse Fourier Transformation (Luu et al., 2014).

2.3. Model evaluation

The trained model was evaluated by the data from the testing dataset, which were new to the model. The linear fit method (LFM) (Iosa et al., 2014) was implemented to assess the similarity between the actual and predicted lower-limb joint angle time series. Three LFM parameters, including the angular coefficient (a_1), the intercept of the fitting line (a_0), and coefficient of determination (R^2) were calculated according to the method introduced in Iosa et al. (2014). The angular coefficient measures the mean variation of the predicted result for every one-unit change in the actual joint angle time series. The intercept of the fitting line measures the shift between the actual and predicted lower-limb joint angle time series. In our study, the absolute value of the a_0 was recorded ($|a_0|$). The coefficient of determination measures the strength of the

linear relationship between the actual and predicted lower-limb joint angle time series. The root mean squared errors (RMSE) were also calculated since RMSEs were often reported in previous gait pattern prediction studies. Furthermore, to access the LASSO Shrinkage performance on prediction variable reduction, the resulted β^{LASSO} after the LASSO shrinkage were reported.

Two previously reported gait pattern prediction models, i.e., the support vector regression (SVR) model and the Gaussian process regression (GPR) model, were applied with the same training and testing datasets as in our proposed LASSO regression model, and their prediction performance was compared with our model. The SVR model was realized by implementing *svm.SVR* class function from Scikit-learn Python module (v0.21.2), and the radial basis function (RBF) was selected as the kernel function as it was reported to be superior to linear and polynomial kernel functions in gait pattern prediction applications (Kim et al., 2009). The GPR model was realized by implementing *gaussian_process.GaussianProcessRegressor* class function from Scikit-learn Python module (v0.21.2). The covariance function of the GPR model was designed to have a composition of the constant kernel adding with the RBF kernel, as suggested in Rasmussen and Williams (2006). The reason that we chose the SVR model and the GPR model for comparisons is because these two models have been widely applied in gait pattern

Table 2
Model evaluation results: Mean (SD).

Cadence	Hip				Knee				Ankle			
	a_1	$ a_0 $ (degree)	R^2	RMSE (degree)	a_1	$ a_0 $ (degree)	R^2	RMSE (degree)	a_1	$ a_0 $ (degree)	R^2	RMSE (degree)
36 step/min	0.98 (0.03)	3.55(1.42)	0.95 (0.03)	3.57(1.43)	0.91 (0.03)	5.46(1.42)	0.94 (0.03)	4.54(1.69)	0.92 (0.03)	3.34(1.24)	0.68 (0.05)	3.68(1.24)
48 step/min	0.96 (0.04)	3.47(1.24)	0.93 (0.02)	3.41(1.49)	0.92 (0.04)	5.76(1.06)	0.97 (0.02)	4.51(1.71)	0.94 (0.04)	3.47(1.35)	0.74 (0.05)	3.62(1.40)
60 step/min	0.98 (0.03)	3.58(1.31)	0.93 (0.02)	3.48(1.48)	0.92 (0.04)	5.32(1.24)	0.96 (0.02)	4.47(1.58)	0.94 (0.04)	3.41(1.33)	0.77 (0.04)	3.49(1.30)
72 step/min	0.97 (0.03)	3.56(1.45)	0.92 (0.02)	3.52(1.53)	0.91 (0.04)	4.71(1.16)	0.98 (0.03)	4.47(1.56)	0.93 (0.05)	3.55(1.34)	0.80 (0.04)	3.54(1.35)
84 step/min	0.98 (0.02)	3.52(1.43)	0.90 (0.02)	3.52(1.62)	0.90 (0.03)	5.05(1.21)	0.97 (0.03)	4.47(1.60)	0.95 (0.04)	3.64(1.33)	0.83 (0.03)	3.45(1.35)
96 step/min	0.98 (0.03)	3.54(1.42)	0.92 (0.01)	3.52(1.53)	0.90 (0.04)	5.01(1.35)	0.97 (0.03)	4.55(1.58)	0.93 (0.04)	3.33(1.42)	0.83 (0.04)	3.58(1.38)

Table 3
Model coefficients (β^{LASSO}) reduction after the LASSO shrinkage.

	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	X15	X16
Hip Joint																
36 step/min								×	×	×		×			×	×
48 step/min						×		×	×			×				×
60 step/min						×		×	×			×	×		×	
72 step/min						×		×	×			×	×			
84 step/min						×		×	×			×				×
96 step/min						×		×	×			×	×			×
Knee Joint																
36 step/min							×						×			
48 step/min			×	×				×								
60 step/min				×			×				×					
72 step/min	×		×	×			×	×		×	×					
84 step/min			×	×	×		×	×	×		×				×	
96 step/min				×	×			×	×						×	
Ankle Joint																
36 step/min			×	×			×	×	×				×	×		×
48 step/min			×	×	×			×	×	×				×		×
60 step/min			×	×	×			×	×			×		×		×
72 step/min			×	×	×				×			×		×		×
84 step/min			×	×	×				×			×		×		×
96 step/min			×	×			×		×					×		×

'×' indicates the corresponding predication variable's coefficient for the specification of Fourier coefficient vector was reduced to zero after the LASSO shrinkage. The definitions of prediction variables (X1 to X16) are given in Table 1.

Table 4

Mean (SD) values of the LFM parameters and RMSE for the LASSO regression model, the GRR model, and the SVR model.

Model	Hip				Knee				Ankle			
	a_1	$ a_0 $ (degree)	R^2	RMSE (degree)	a_1	$ a_0 $ (degree)	R^2	RMSE (degree)	a_1	$ a_0 $ (degree)	R^2	RMSE (degree)
LASSO	0.98 (0.03)	3.54(1.21)	0.96 (0.05)	3.50(1.51)	0.92 (0.03)	5.22(1.21)	0.96 (0.03)	4.50(1.62)	0.94 (0.03)	3.45(1.01)	0.78 (0.03)	3.56(1.34)
GPR	0.97 (0.04)	3.44(1.66)	0.96 (0.03)	3.69(1.64)	0.92 (0.04)	4.91(0.99)	0.98 (0.03)	4.35(2.42)	0.91 (0.04)	3.99(1.20)	0.78 (0.06)	4.26(2.31)
SVR	0.97 (0.02)	3.56(1.41)	0.98 (0.02)	3.59(1.72)	0.91 (0.04)	5.42(1.05)	0.96 (0.02)	4.61(2.08)	0.93 (0.05)	3.54(1.11)	0.79 (0.04)	3.56(1.89)

prediction applications (Kim et al., 2009; Yun et al., 2014; Wu et al., 2018), and are not much dependent on the large amount of training data just like our proposed model.

3. Results

Table 2 shows the mean (SD) values of the LFM parameters and RMSE between the actual and predicted joint angle trajectories under different walking cadence conditions for the proposed LASSO regression model. The angular coefficient (a_1) ranged from 0.96 to 0.98, 0.90–0.92, and 0.92–0.95, for the hip, knee and ankle, respectively. The intercept of the fitting line ($|a_0|$) ranged from 3.47° – 3.58° , 4.71° – 5.76° , and 3.33° – 3.64° , for the hip, knee and ankle, respectively. Coefficient of determination (R^2) ranged from 0.90 to 0.95, 0.94–0.98, and 0.68–0.83, for the hip, knee, and ankle, respectively. The RMSE for the hip, knee and ankle across different cadence levels ranged from 3.41° – 3.57° , 4.47° – 4.55° , and 3.45° – 3.68° , respectively. Table 3 shows the results of LASSO shrinkage for each joint angle trajectory. We noted that the model coefficients of some prediction variables were set to zero after LASSO shrinkage. The final gait pattern prediction models varied across joints and cadence levels.

The results for the comparisons of the LFM parameters and RMSE among the LASSO regression model, the SVR model, and the GRP model are given in Table 4. On average, the LASSO regression model had the largest a_1 (0.98), and smallest RMSE (3.50°) for the hip joint. For the ankle joint, the LASSO regression model gave the largest a_1 (0.94), smallest $|a_0|$ (3.45°), and smallest RMSE (3.56° , same as SVR). For the knee joint, both LASSO and GPR model had the largest a_1 (0.92). A randomly selected example of the joint angle prediction by the three models was illustrated in Fig. 1.

4. Discussion

This study presents a novel individualized gait pattern prediction model based on the LASSO regression. The results show that the RMSEs between the actual and predicted lower-limb joint angle time series were all smaller than five degrees (Table 2). Further, the LFM parameters were used to reveal the waveform similarity between the actual and predicted lower-limb joint angle time series. Specifically, the resulted a_1 at different cadence levels was close to one, suggesting small amplitude differences between the actual and predicted joint angle time series. The resulted $|a_0|$ was all under six degrees for the lower-limb joints. This indicated that the predicted joint angles had acceptable offsets in magnitude compared to actual joint angles, given that the ranges of motion of the lower-limb joint angles were approximately 60° . These results suggest that our proposed model can accurately predict the lower-limb joint angle trajectories during gait.

Some previous studies reported smaller RMSEs. For example, Koopman et al. (2014) achieved the RMSEs ranging between 2.33° and 4.85° , and Luu et al. (2014) reported RMSEs ranging

between 2.13° and 3.46° . It is worth noting that these previous studies mainly used cross-validation to evaluate their models. The cross-validation is an iterative process, where the data are iteratively used in both model training and model evaluation. Though the prediction errors are typically smaller with the cross-validation (Halilaj et al., 2018), it sacrifices the external validity of the model. In the present study, instead of using cross-validation, the proposed model was evaluated by using a novel testing dataset that had not been used in model training. Therefore, the evaluation results can be well generalized into new scenarios and external validity of the proposed model can be ensured.

When compared with the previously reported gait pattern prediction models (i.e., GRP model and SVR model) with the same training and testing datasets, the prediction performance of our proposed model was comparable to them (Table 4). However, the number of prediction variables was smaller in our proposed model due to the shrinkage process. This can help reduce model variance and enhance model's generalizability (Ferber et al., 2016).

Results from the shrinkage process (Table 3) could provide insights into the contributions of the selected prediction variables (Table 1) to gait pattern planning. For example, the coefficient of malleolus height were all set to zero after shrinkage process for the ankle joint, suggesting that malleolus height might not be closely related with ankle joint planning. Instead, malleolus height could play a more important role in hip joint and knee joint planning, as it remained as a predictor for hip joint and knee joint after the shrinkage process. It is also worth noting that the roles of the selected prediction variables in gait pattern planning were dependent on the cadence level. Similar findings were also reported in previous studies. Wu et al. (2018), for instance, found that the weightings of body parameters changed with walking speed in their GRP based gait pattern prediction model.

Some previous studies took walking speed as a prediction variable in their gait pattern prediction model (Yun et al., 2014; Wu et al., 2018). However, in the present study, the walking speed was regulated at six different cadence levels, and different LASSO regression based models were developed to predict the joint trajectories under different cadence levels. We did so because of some practical considerations. In particular, robot-assisted rehabilitation typically requires the walking speed controlled and specified by the users or training professionals according to their needs at different stages of the training and rehabilitation process.

With patients' age, body parameters and target cadence as the inputs, the proposed LASSO regression model can generate individualized gait patterns and provide a better exoskeleton trajectory planning. This can facilitate the design of the individual adaptable gait rehabilitation exoskeleton, which will lead to improved motor learning and result in better therapeutic outcomes (Werner et al., 2002).

This study has some limitations. Firstly, the data used for model training and evaluation were collected from young participants. Although age was considered as a prediction variable, the range of this variable was limited (18–31 years old). Future work should

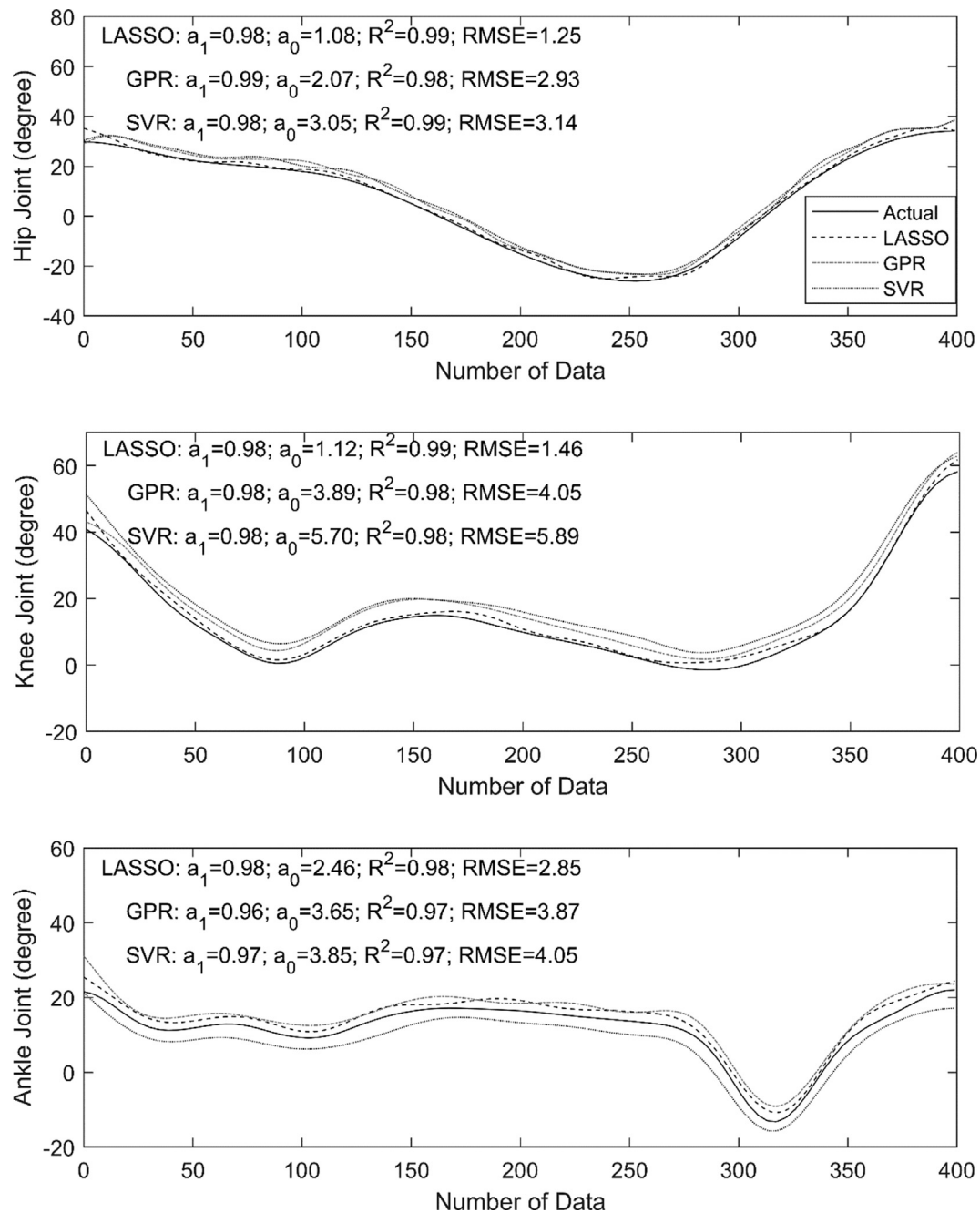


Fig. 1. A randomly selected example of the joint angle prediction by the LASSO, GPR, and SVR models with cadence = 60 step/min.

include older participants in model development and evaluation. Secondly, although the LASSO regression model automatically selected and reduced the number of prediction variables based on its model fitting procedure, it is still difficult to explain the potential collinearity between the prediction variables. Therefore, future research needs to explicitly investigate such collinearity and their effects on individualized the gait pattern. In addition, although the results from the shrinkage process may help better understand the relationship between the selected prediction variables and gait pattern planning, the proposed model in essence is a statistical model and the underlying biological mechanisms for such relationship is still unknown and worth further investigation.

In summary, this paper presents a novel individualized gait pattern prediction model based on the LASSO regression. The evaluation results have demonstrated that our model can accurately

predict gait patterns. To our best knowledge, the present study is the first attempt to implement the LASSO regression in gait pattern prediction. This model can benefit exoskeleton trajectory planning. With age, gender and body anthropometric data as the prediction variables, this model can address the individual differences, and provide more comfortable and safer interaction experience for users.

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Declaration of Competing Interest

None to declare.

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